Course Code: CSE3206
Course Title: Computer Networks Lab
Submitted To: Md. Kaviul Hossain
Submitted By: Anika Raisa Hridi
ID: 233014028

1. Introduction

This report presents the design and implementation of a **Wide Area Network (WAN)** using Cisco Packet Tracer. The network consists of multiple Local Area Networks (LANs) interconnected through routers.

The primary objective is to enable seamless communication between all devices using the **Routing Information Protocol (RIP v2)** dynamic routing protocol.

2. Objectives

- To design a WAN topology with routers, switches, PCs, and servers
- To configure IP addressing using /24 subnetting
- To implement dynamic routing using RIP v2
- To verify end-to-end connectivity between all networks

3. Network Topology Description

The topology consists of:

- **3 Routers**
- **3 Switches**
- **5 PCs**
- **3 Servers**

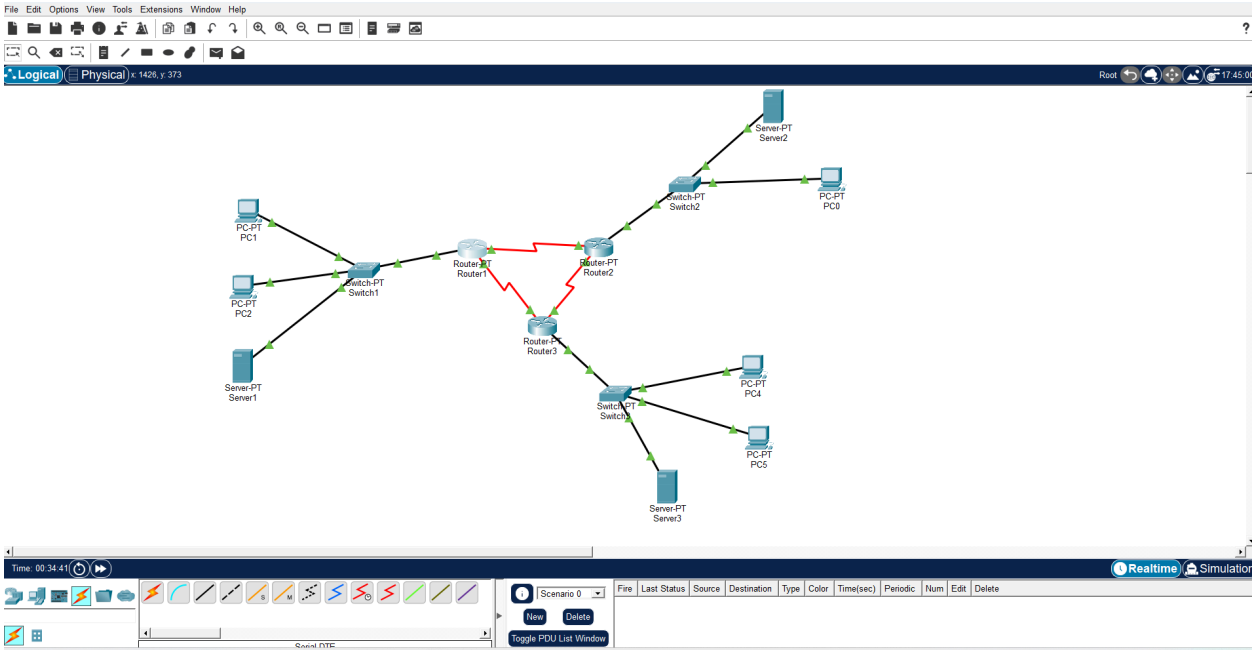
Three separate LANs are connected through WAN links between routers.

Network Segments:

- LAN 1: 192.168.1.0/24
- LAN 2: 192.168.2.0/24
- LAN 3: 192.168.3.0/24

WAN Links:

- R1 – R2: 175.1.10.0/24
- R1 – R3: 175.1.11.0/24
- R2 – R3: 175.1.12.0/24



4. IP Addressing Scheme

4.1 LAN 1 (Left Network)

Device	IP Address
PC1	192.168.1.2
PC2	192.168.1.3
Server1	192.168.1.4
Router1	192.168.1.1

4.2 LAN 2 (Top Right Network)

Device	IP Address
PC0	192.168.2.3
Server2	192.168.2.2
Router2	192.168.2.1

4.3 LAN 3 (Bottom Right Network)

Device	IP Address
PC4	192.168.3.2
PC5	192.168.3.3
Server3	192.168.3.4
Router3	192.168.3.1

4.4 Router-to-Router Links

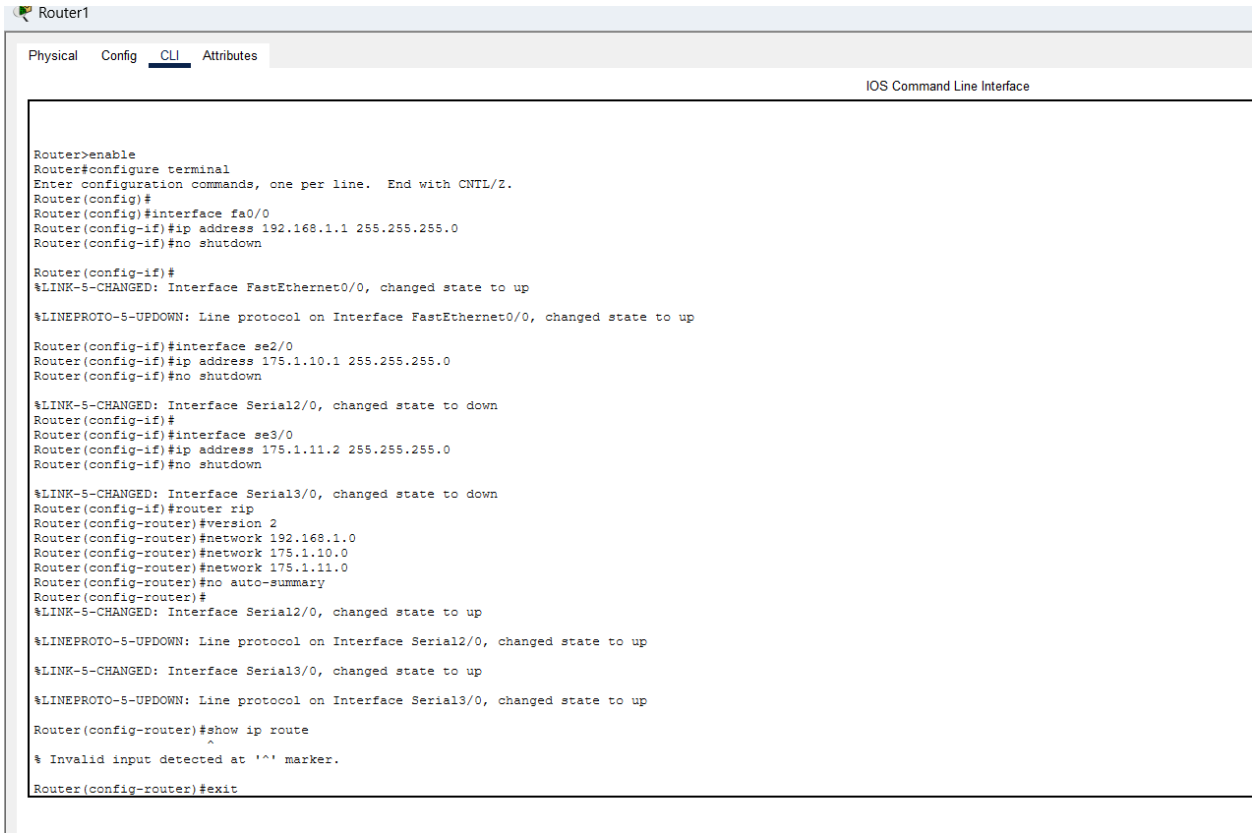
Link	IP Assignment
R1 – R2	175.1.10.1 / 175.1.10.2
R1 – R3	175.1.11.2 / 175.1.11.1
R2 – R3	175.1.12.2 / 175.1.12.1

5. Configuration Methodology

5.1 Router Configuration

Each router interface was configured with appropriate IP addresses and activated using the `no shutdown` command.

RIP version 2 was enabled on all routers with the relevant network statements. Auto-summary was disabled to ensure proper classless routing.




```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#interface fa0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#interface se2/0
Router(config-if)#ip address 175.1.10.1 255.255.255.0
Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial2/0, changed state to down
Router(config-if)#
Router(config-if)#interface se3/0
Router(config-if)#ip address 175.1.11.2 255.255.255.0
Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial3/0, changed state to down
Router(config-if)#router rip
Router(config-router)#version 2
Router(config-router)#network 192.168.1.0
Router(config-router)#network 175.1.10.0
Router(config-router)#network 175.1.11.0
Router(config-router)#no auto-summary
Router(config-router)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up

%LINK-5-CHANGED: Interface Serial3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial3/0, changed state to up

Router(config-router)#show ip route
                     ^
% Invalid input detected at '^' marker.

Router(config-router)#exit
```

Router2

Physical Config CLI Attributes

IOS Command Line Interface

```
Press RETURN to get started!



Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface fa0/0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#interface se2/0
Router(config-if)#ip address 175.1.10.2 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up

Router(config-if)#interface se3/0
Router(config-if)#ip address 175.1.12.2 255.255.255.0
Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial3/0, changed state to down
Router(config-if)#router rip
Router(config-router)#version 2
Router(config-router)#network 192.168.2.0
Router(config-router)#network 175.1.10.0
Router(config-router)#network 175.1.12.0
Router(config-router)#no auto-summary
Router(config-router)#
%LINK-5-CHANGED: Interface Serial3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial3/0, changed state to up
```

Router3

Physical Config CLI Attributes

IOS Command Line Interfac

```
Press RETURN to get started!



Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface fa0/0
Router(config-if)#ip address 192.168.3.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#interface se2/0
Router(config-if)#ip address 175.1.11.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up

Router(config-if)#interface se3/0
Router(config-if)#ip address 175.1.12.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
Router(config-if)#router rip
Router(config-router)#version 2
Router(config-router)#network 192.168.3.0
Router(config-router)#network 175.1.11.0
Router(config-router)#network 175.1.12.0
Router(config-router)#no auto-summary
%LINK-5-CHANGED: Interface Serial3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial3/0, changed state to up
```

5.2 End Device Configuration

All PCs and servers were configured with:

- IP Address (as per table)
- Subnet Mask: 255.255.255.0
- Default Gateway corresponding to their router interface

6. Routing Protocol Implementation

The **Routing Information Protocol** was used as the dynamic routing protocol.

Key features implemented:

- Version 2 (classless routing support)
- Network advertisement for all connected networks
- Automatic route exchange between routers

```
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     175.1.0.0/24 is subnetted, 3 subnets
C       175.1.10.0 is directly connected, Serial2/0
C       175.1.11.0 is directly connected, Serial3/0
R       175.1.12.0 [120/1] via 175.1.10.2, 00:00:15, Serial2/0
                   [120/1] via 175.1.11.1, 00:00:05, Serial3/0
C    192.168.1.0/24 is directly connected, FastEthernet0/0
R    192.168.2.0/24 [120/1] via 175.1.10.2, 00:00:15, Serial2/0
R    192.168.3.0/24 [120/1] via 175.1.11.1, 00:00:05, Serial3/0

Router#
```

7. Testing and Verification

To verify connectivity, multiple **ICMP ping tests** were performed between devices across different networks.

Sample Tests:

- PC1 → PC4
- PC2 → Server2
- PC5 → Server1

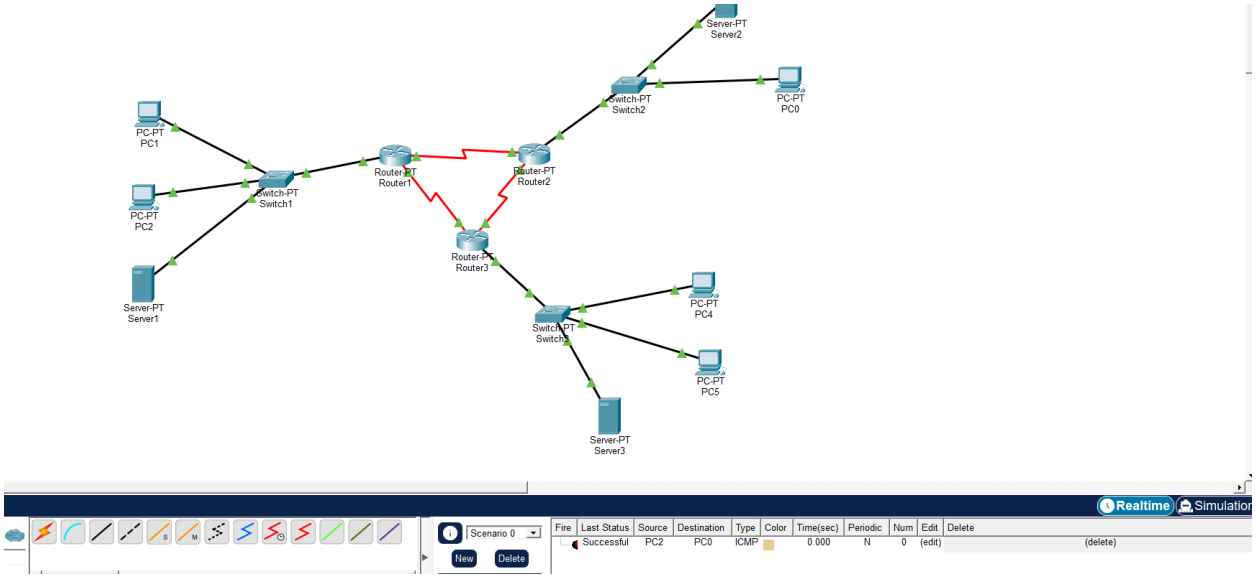
All tests were successful, indicating proper routing and connectivity.

```
C:\>ping 192.168.3.2

Pinging 192.168.3.2 with 32 bytes of data:

Reply from 192.168.3.2: bytes=32 time=14ms TTL=126
Reply from 192.168.3.2: bytes=32 time=5ms TTL=126
Reply from 192.168.3.2: bytes=32 time=1ms TTL=126
Reply from 192.168.3.2: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.3.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 14ms, Average = 5ms

C:\>
```

8. Simulation Analysis

The simulation mode in Cisco Packet Tracer was used to analyze packet flow.

- Simple PDUs were created
- Packets successfully traversed across routers
- No packet loss was observed



9. Results

- All routers successfully exchanged routing information
- Routing tables contained all remote networks
- Full connectivity was achieved between all LANs
- Network performed as expected under RIP v2

10. Conclusion

The WAN topology was successfully designed and implemented. The use of **dynamic routing (RIP v2)** ensured efficient and automatic route discovery between networks.

All devices across different LANs were able to communicate reliably, demonstrating the correctness of the configuration and overall network design.